

Short notes

Understand the idea. Remember the essentials. Check the source.

01 Electric charge

Key concept

Electric charge is a property of matter that causes electrical interactions.

 p. 1 SI

Check your understanding

What happens when two positive charges are brought near each other?

 p. 1 SI

Must-know points

- Two types: positive and negative.

 p. 1 SI

- Like charges repel; unlike charges attract.

 p. 1 SI

- SI unit: coulomb (C).

 p. 1 SI

- Conserved in an isolated system; quantised in multiples of e .

 p. 1 SI

 p. 1 SI

 Key concept

Coulomb's law gives the electrostatic force between two stationary point charges.

 p. 2 S2

 Check your understanding

What happens to force when separation doubles?

 p. 2 S2

Must-know points

- $F = k |q_1 q_2| / r^2$.

 p. 2 S2

- Force acts along the line joining the charges.

 p. 2 S2

- Double $r \rightarrow$ force becomes $F/4$.

 p. 2 S2

- Net force is the vector sum of individual forces.

 p. 2 S2

 Formula & units

$F = k |q_1 q_2| / r^2$; $k = 1/(4\pi\epsilon_0)$ in vacuum.

 p. 2 S2

 Example from the evidence

If separation changes from r to $2r$, the force changes from F to $F/4$.

 p. 2 S2

03 Electric field

Key concept

Electric field is the force per unit positive test charge:

$$E = F/q.$$

 p. 3 S3

Check your understanding

Why should a test charge be small?

 p. 3 S3

Must-know points

- Vector quantity: magnitude and direction.

 p. 3 S3

- SI unit: N/C.

 p. 3 S3

- Use a small test charge to avoid disturbing source charges.

 p. 3 S3

- Field lines go away from positive charge and toward negative charge; they do not intersect.

 p. 3 S3

Formula & units

$$E = F/q; \text{ unit} = \text{N/C}.$$

 p. 3 S3

 p. 3 S3

04 Electric potential

Key concept

Electric potential is potential energy per unit charge: $V = U/q$.

 p. 4 S4

Avoid this confusion

Incorrect: zero potential always means zero field. Correct: a field can exist where potential is zero.

 p. 4 S4

Must-know points

- Scalar quantity: no vector direction.

 p. 4 S4

- SI unit: volt = joule/coulomb.

 p. 4 S4

- Potential difference is change in potential energy per unit charge between two points.

 p. 4 S4

- Zero potential does not necessarily mean zero electric field.

 p. 4 S4

Check your understanding

Is electric potential a vector?

 p. 4 S4

Formula & units

$V = U/q$; unit = volt = J/C.

 p. 4 S4

 p. 4 S4